Question #1 of 32

Question ID: 1473556

Credit valuation adjustment is *most likely*:

A) higher when the recovery rate is higher.

B) the sum of present values of expected losses.

C) higher when the probability of survival is higher.

Explanation

Credit valuation adjustment (CVA) is the sum of present values of expected losses. CVA is positively related to the probability of default and negatively related to probability of survival and recovery rate.

(Module 28.1, LOS 28.a)

Question #2 of 32

Question ID: 1473569

To analyze the credit risk of a company with significant off-balance sheet liabilities, which credit model is *most appropriate*?

A) Structural model.

B) Reduced form model.

C) Econometric model.

Explanation

Structural models are not suitable when the company has complex balance sheets or when there are significant off-balance sheet liabilities. Reduced form models would be appropriate in such a situation.

(Module 28.4, LOS 28.d)

Question #3 of 32

Question ID: 1473564

Fico scores are inversely related to the:

A) variety of credit types used.

B) number of 'hard' inquiries.

C) length of credit history.

Explanation

FICO scores are higher for those with: (a) longer credit histories (age of oldest account), (b) absence of delinquencies, (c) lower utilization (outstanding balance divided by available line), (d) fewer credit inquires, and (e) a variety of types of credit used.

(Module 28.3, LOS 28.b)

Question #4 of 32

Question ID: 1480345

Which of the following statements regarding evaluating credit risk of Asset Backed Securities (ABS) is *least* accurate?

A) The analysis should entail consideration of the composition of the collateral pool and the cash flow waterfall.

B) Unlike for corporate debt, structural and reduced form models are not appropriate.

C) Credit rating agencies do not use the same credit ratings for ABS as for corporate debt.

Explanation

Reduced form and structural models can be used as long as they take into account the complex structure of the ABS. Secured debt is usually financed via a bankruptcy-remote SPE. This isolation of securitized assets allows for higher credit rating and lower cost to the issuer.

(Module 28.7, LOS 28.h)

Question #5 of 32

Question ID: 1473570

Which key input into a reduced form model can be estimated using a regression model?

A) Default intensity.

B) Loss intensity.

C) Recovery rate.

Explanation

Default intensity is the probability of default over the next time period and can be estimated using regression models.

(Module 28.4, LOS 28.d)

Question #6 of 32

Question ID: 1473566

Under the structural model, owning equity in a company is equivalent to:

A) long position in a call option on the assets of the company. ✓

B) short position in a put option on the assets of the company. ✗

C) long position in a call option on the firm's debt. ✗

Explanation

Equity investors have economic position equivalent to a long position in a call option on the assets of the company with a strike price equal to the face value of debt.

(Module 28.4, LOS 28.d)

Question #7 of 32

Question ID: 1473559

Perez Zinta has collected the following information on a 3-year, 3% corporate bond.

Year	Exposure	LGD	PD	PS	Expected Loss	DF	PV of Expected Loss
1	103.96	41.585	1.80%	98.200%	0.749	0.9756	0.73
2	103.49	41.395	1.77%	96.432%	0.732	0.9518	0.70
3	103.00	41.200	1.74%	94.697%	0.715	0.9286	0.66
						CVA	**2.091**

Given a 3-year risk-free rate of 1.50%, Calculate the IRR of the bond assuming that default occurs in year 2.

A) -13.37% ✗

B) -20.60% ✓

C) -25.48% ✗

Explanation

First calculate the VND: N=3, PMT = 3, FV = 100, I/Y = 1.50, PV = 104.37 = VND.

Price of the corporate bond = VND – CVA = 104.37 – 2.09 = 102.28

Cash flow in year 0 = -102.28, cash flow in year 1 = $3 (coupon, no default).

If the bond defaults in year 2, recovery = Exposure – LGD = 103.49 – 41.40 = 62.09 = cash flow in year 2.

Enter the cash flows and calculate IRR = -20.60%.

(Module 28.2, LOS 28.a)

Question #8 of 32

Question ID: 1473584

As compared to otherwise identical corporate debt, securitized debt is *least likely* to have:

A) higher leverage for the issuer.

B) lower cost for the issuer.

C) the same risk premium.

Explanation

The isolated structure of securitized assets allows for higher leverage and lower cost to the issuer. Investors also benefit from greater diversification, more stable cash flows and a *higher* risk premium relative to similar rated general obligation bonds (due to higher complexity associated with collateralized debt).

(Module 28.7, LOS 28.h)

Question #9 of 32

Question ID: 1473580

Which of the following two securities are *most likely* used to calculate the term structure of credit spreads?

A) A corporate issuer's zero coupon bond and a default free zero coupon bond.

B) A corporate issuer's senior debt and the same issuer's subordinated debt.

C) A corporate issuer's coupon paying bond and the same issuer's zero coupon bond.

Explanation

If a zero coupon bond is not available an implied zero coupon bond price for the issuer can be derived from the coupon paying bond price.

(Module 28.6, LOS 28.g)

Question #10 of 32

Question ID: 1473586

An investor in an ABS would face which risks on account of the ABS servicer?

A) Operational and counterparty risk. ✓

B) Credit and concentration risk. ✗

C) Operational and concentration risk. ✗

Explanation

After origination, investors in secured debt face the operational and counterparty risk of the servicer.

(Module 28.7, LOS 28.h)

Question #11 of 32

Question ID: 1473571

When assessing a company's credit risk using structural models, which of the following statements is *most* accurate?

A) Structural models do not account for the impact of interest rate risk of the value of a company's assets. ✓

B) Owning equity is economically equivalent to owning a risk free bond and simultaneously selling a put option on the assets of the company. ✗

C) Owning debt is economically equivalent to owning a European call option on the company's assets. ✗

Explanation

Owning equity is economically equivalent to owning a European call option on the assets of the company. Owning debt is economically equivalent to owning a risk free bond and simultaneously selling a put option on the assets of the company. The structural model assumes that risk-free rate is not stochastic (i.e., it assumes that risk-free rate is constant).

(Module 28.4, LOS 28.d)

Question #12 of 32

Question ID: 1473585

An ABS security backed by a highly granular collateral pool composed of hundreds of clearly defined loans, analysis of collateral pool can be done using:

A) distribution waterfall analysis.

B) summary statistics for analyzing credit risk.

C) examination of individual loans.

Explanation

A highly granular pool would have hundreds of clearly defined loans, allowing for use of summary statistics as opposed to investigating each borrower. A more-discrete pool of few loans would warrant examination of each obligation separately. Distribution waterfall analysis is part of evaluation of the ABS structure (and not collateral pool).

(Module 28.7, LOS 28.h)

Question #13 of 32

Question ID: 1473561

Credit scores are *most likely* to be used for:

A) sovereign bonds.

B) ABS.

C) small businesses.

Explanation

Credit scores are used for individuals and small businesses. Credit ratings are used for corporate, quasi-government, and sovereign bonds as well as for secured debt (ABS).

(Module 28.3, LOS 28.b)

Question #14 of 32

Question ID: 1473587

As compared to other secured debt, investors in a covered bond have:

A) an embedded conversion option.

B) an embedded put option.

C) recourse rights.

Explanation

Covered bonds are backed by the collateral pool as well as by the issuer; investors in covered bonds have recourse rights.

(Module 28.7, LOS 28.h)

Question #15 of 32

Question ID: 1473565

Mihor Kotak is evaluating the impact of a ratings upgrade on 1Team bonds. The bonds have a modified duration of 5.88 and the current credit spread on the bonds is 60 bps. After the upgrade, Kotak expects that the spreads will narrow by 15bps. Based on Kotak's expectations, what will be the estimated change in the price of the bond if the upgrade occurs?

A) 8.82%

B) 0.88%

C) 0.38%

Explanation

Change in spread (given) = – 15 bps

$\Delta\%P = -$ (modified duration of the bond) × (Δ spread) = $-5.88 \times -0.0015 = -0.00882$ or 0.88%. Since spread narrows, price will increase (i.e., a positive price change).

(Module 28.3, LOS 28.c)

Question #16 of 32

Question ID: 1473563

Higher rated bonds have lower:

A) returns.

B) price.

C) credit spreads.

Explanation

Higher rated bonds have lower spreads. Price and return depends on other factors (e.g., coupon rate, maturity, risk-free rate).

(Module 28.3, LOS 28.b)

Question #17 of 32

Question ID: 1473555

A corporate bond has one year to maturity with a probability of default of 2.05% and a recovery rate of $32.00 per $100 par value. If an investor holds $100,000 of par value, what is the expected loss?

A) $656.

B) $1,394.

C) $2,050.

Explanation

Expected loss	= Probability of default × expected loss per $ × par value
	= 0.0205 × (1 − 0.32) × $100,000
	= $1,394

(Module 28.1, LOS 28.a)

Question #18 of 32

Question ID: 1473568

Using the structural model, the value of the put option on the assets of the company is equal to:

A) credit valuation adjustment of the bond.

B) value of the risky bond minus value of the risk-free bond.

C) the value of the call option on assets of the company.

Explanation

Under structural model the put option value = value of risk-free bond – value of the risky bond = CVA.

(Module 28.4, LOS 28.d)

Question #19 of 32

Question ID: 1473579

If investors are expecting an impending recession, credit spreads would *most likely*:

A) remain unchanged.

B) widen.

C) narrow.

Explanation

Credit spreads change based on market's expectations. Impending recessions would lead to upward revision in probability of default and lower recovery rate. Combined, these revisions would lead to widening of credit spreads.

(Module 28.6, LOS 28.f)

Freeman LLC, is a large investment firm based on the East Coast of the United States. The company manages a range of investment funds with several different objectives but focuses mainly on fixed income investments. Josh Scowen is a credit analyst who has just taken up a position with the firm and is currently familiarizing himself with the various models and techniques used by Freeman.

Scowen's first task is to assess the present value of the expected loss (CVA) on a bond issued by Dreamy, Inc., an online retailer of designer fashion products. The company expanded rapidly two years ago, but business conditions have deteriorated recently. Scowen's supervisor is concerned that the company may run into serious trouble soon.

Exhibit 1: Dreamy Bond

Par	\$1,000
Annual coupon	8%
Time to maturity	2 years

Note: the risk-free rate of return is 1.22% (assume a flat yield curve).

Freeman also makes extensive use of reduced form and structural models to assess credit risk. Scowen's supervisor has asked him to review the details of the approaches Freeman uses.

Scowen recalls using a reduced form model at a previous firm and believes that the following three assumptions are valid:

Assumption 1:	The company's liabilities can be modelled as a single zero-coupon bond.
Assumption 2:	The risk-free interest rate is constant.

Assumption 3:	The probability of default and the recovery rate are not constant.

Freeman has recently used a reduced form model to analyze the credit risk of a zero-coupon bond issued by Sleepy, Inc. Exhibit 2 lists some of the details of the simple reduced form model.

Exhibit 2: Sleepy Bond, Reduced Form Model

Coupon:	Zero
Face value:	$10,000
Time to maturity:	1 year
Hazard rate:	0.02
Loss given default:	35%
One-year, default-free, zero-coupon bond price ($1 par):	0.95
Credit valuation adjustment:	66.50

Scowen also has a background in option pricing theory from a previous role and is confident that he can put this experience to good use when using a structural model. He believes that structural models value risky debt of a company by deducting the value of a put option on a company's assets from the value of otherwise identical risk-free debt.

Question #20 - 23 of 32

Question ID: 1473574

Using the information in Exhibit 1, the expected exposure after one year is *closest* to:

A) $1,146.98.

B) $1,023.76.

C) $1,066.98.

Explanation

The expected exposure is the present value (@ risk-free rate of 1.22%) of the remaining cash flows on the bond. After one year, the remaining cash flow on the bond is the currently due coupon payment of $80 (issuer would not default *after* paying the coupon) plus the last coupon plus principal of $1080.

expected exposure = 80 + $1,080 / 1.0122 = $1,146.98.

(Module 28.5, LOS 28.e)

Question #21 - 23 of 32

Question ID: 1473575

Which of the assumptions stated by Scowen regarding the reduced form model is *most* accurate?

A) Assumption 3.

B) Assumption 2.

C) Assumption 1.

Explanation

The assumptions of reduced form models include:

- The risk-free interest rate is stochastic.
- The state of the economy is stochastic and depends on macroeconomic variables.
- The probability of default (default intensity) and the recovery rate depend on the state of the economy and are not constant.

(Module 28.1, LOS 28.a)

Question #22 - 23 of 32

Question ID: 1473576

Using information in Exhibit 2, the value of the Sleepy Bond is *closest* to:

A) $9,566.27.

B) $9,433.50.

C) $9,500.00.

Explanation

The credit valuation adjustment is $66.50, which represents the difference between the price of a risky bond and the equivalent risk-free bond. The one-year risk-free bond price is $9,500 (for a $10,000 par value).

bond value = 10,000 – 66.50 = $9,433.50.

(Module 28.4, LOS 28.d)

Question #23 - 23 of 32

Question ID: 1473577

Scowen's comment regarding option pricing theory and structural models is *best* described as:

A) accurate. ✓

B) inaccurate, as structural models value risky debt by adding the value of a put option on the company's assets to the value of risk-free debt. ✗

C) inaccurate, as structural models value risky debt by deducting the value of a call option on the company's assets from the value of risk free debt. ✗

Explanation

Scowen's statement is correct. Under structural models:

value of risky debt = value of risk-free debt – value of put option on company assets

(Module 28.4, LOS 28.d)

Question #24 of 32

Question ID: 1473581

Which of the following factors is *least likely* a determinant of term structure of credit spreads?

A) Equity market volatility. ✗

B) Financial conditions in the market. ✗

C) Existence of off-balance sheet liabilities. ✓

Explanation

Term structure of credit spread is influenced by credit quality, financial conditions, market demand and supply, and equity market volatility.

(Module 28.6, LOS 28.g)

Question #25 of 32

Question ID: 1473572

Zack Ma is evaluating a five-year, 4% Zem bond. Ma has calculated the CVA on the bond to be $2.12 per $100 par. Current benchmark rates are flat at 3%. The credit spread on the bond is *closest* to:

A) 0.46%

B) 0.97%

C) 0.21%

Explanation

First calculate the VND: N=5, PMT = 4, FV = 100, I/Y = 3. PV = 104.58 = VND.

Value of risky bond = VND – CVA = 104.58 – 2.12 = 102.46

YTM on risky bond: N=5, PV = -102.46, PMT = 4, FV = 100, I/Y = 3.46%

Credit spread = YTM (risky) – YTM (risk-free) = 3.46% – 3% = 0.46%.

(Module 28.5, LOS 28.e)

Question #26 of 32

Question ID: 1473558

If the annual hazard rate for a bond is 1.80%, the probability that the bond does not default over the next three years is *closest* to:

A) 96.30%

B) 95.20%

C) 94.70%

Explanation

Probability of survival = $(1 – 0.018)^3$ = 0.9470.

(Module 28.1, LOS 28.a)

Question #27 of 32

Question ID: 1473578

Zack Ma is evaluating a 10-year, 4% Tesa bond. Ma has calculated the CVA on the bond to be $1.19 per $100 par. Ma is considering the impact of a new patent granted to Tesa. After careful analysis, Ma concludes that the probability of default would most likely decrease on the bond. After incorporating the revised probability in his analysis, Ma will *most likely* conclude that:

A) only the credit spread will be lower; the impact on CVA will depend on changes in benchmark rates.

B) both the CVA and the credit spread will be lower.

C) both the CVA and the credit spread will be higher.

Explanation

CVA and credit spreads are positively related to probability of default.

(Module 28.6, LOS 28.f)

Question #28 of 32

Question ID: 1473582

Upward sloping credit curve is *most likely* an indication of:

A) upward sloping benchmark curve.

B) expectations of an economic expansion.

C) expectations of a recession.

Explanation

Upward sloping credit curve indicates widening of spread as debt maturity increases. This would be consistent with expectations of higher probability of default (or lower recovery rate) in the longer-term, which would be consistent with expectations of a recession.

(Module 28.6, LOS 28.g)

Question #29 of 32

Question ID: 1473567

Under the structural model, owning risky debt is equivalent to a long position in a similar risk-free bond and a:

A) short position in a put option on the assets of the company.

B) long position in a call option on the assets of the company.

C) long position in a put option on the assets of the company.

Explanation

Risky debt ownership is economically equivalent to a long position in risk-free bond and a short position in a put option on the assets of the company.

(Module 28.4, LOS 28.d)

Question #30 of 32

Question ID: 1473560

Alan Barding is a bank analyst currently reviewing data on the credit scores of 3 individuals who have applied for a bank loan. The credit scores for the 3 individuals are shown below:

Individual	Credit score
A	700
B	440
C	350

Which of the following conclusions is Barding *least likely* to draw?

A) Individual A has a lower credit risk than individual B.

B) Individual C is twice as likely to default as individual A.

C) Individual B is less likely to default than individual C.

Explanation

Credit scores are ordinal rankings. Individual C is more likely to default than individual A, but it cannot be concluded that A is twice as likely.

(Module 28.3, LOS 28.b)

Question #31 of 32

Question ID: 1473557

Calculate the CVA on a 1.75%, 1-year, $100 par annual pay bond with recovery rate of 70% and probability of default of 2%. Assume that the 1-year risk-free rate is 2%.

A) $1.12

B) $1.89

C) $0.59

Explanation

Year	Exposure	LGD	PD	Expected Loss	DF	PV of Expected Loss
0	99.75	29.93	2.00%	0.60	0.9804	0.59

DF = PV of $1 using risk-free rate = 1 /1.02 = 0.9804. Exposure = 101.75/1.02 = 99.75. LGD = Exposure × (1 – recovery rate) = 99.75 × 0.30. Expected loss = LGD × PD = 29.93 × 0.02.

(Module 28.1, LOS 28.a)

Question #32 of 32

Question ID: 1473562

Credit scores and credit ratings are both:

A) cardinal rankings.

B) ordinal rankings.

C) qualitative ratings.

Explanation

Credit scores and credit ratings are both ordinal rankings.

(Module 28.3, LOS 28.b)